

# Jaehyun Jeon

📍 Seoul, Korea ✉ jaehyunjeon@lgresearch.ai | jaehyunjeon.0126@gmail.com 📞 +82-10-7211-9910

🌐 [Homepage](#) in [LinkedIn](#) [Github](#)

## Experience

### LG AI Research

AI Scientist (EXAONE Lab)

Sep. 2025 – Current

- Contributing to the development of the EXAONE foundation model, focusing on improving Korean language capabilities and knowledge through benchmarks such as the Korean CSAT

### NC AI

Research Engineer (DialogTech Team)

Jan. 2026 – Aug. 2026

- Developed and evaluated domain-specialized AI agents for game customer support, collaborating with support teams to define output schemas, field priorities, and an evaluation dataset
- Built an internal eval and trace visualization platform, designed checklist-based eval metrics, and a configurable experiment system, iterating over tool selection and context compression strategies

### LG AI Research

Research Engineer Internship (Language Lab)

Sep. 2025 – Dec. 2025

- Built ChatEXAONE Extended Thinking, an agentic feature with integrated web search and patent-specialized workflows tailored to legal document structure, trained on QA-derived trajectory data
- Served as internal design manager, ensuring production-ready service behavior working between model development and UX, including streaming integration and citation rendering

### MIR Lab, Yonsei University (Advisor: Youngjae Yu)

Graduate Researcher

Mar. 2024 – Aug. 2025

- Conducted multimodal AI research/projects, focusing on real-world problem solving via model evaluation, alignment, and RAG

Research Internship

Mar. 2023 – Feb. 2024

- Assisted in multimodal AI projects, gaining practical skills in model training/inference at code level and performance analysis

### Data Science Lab (DSL), Yonsei University (Student academic society)

Vice President

Jan. 2022 – Dec. 2022

- Society operations: recruitment, networking, session management, project acquisition & progress tracking
- Managed three industry colab projects: contracts and overall project lead (Freewheelin, Pozalabs, VisualCamp)
- Led sessions/study groups on ML topics (Generative models, NLP, MLOps, etc), and contributed to industry/internal projects

## Education

### Yonsei University

M.S. in Computer Science (GPA: 4.24/4.3, Advisor: Youngjae Yu)

Mar. 2024 – Feb. 2026

B.S. in Computer Science (GPA: 4.26/4.3, Summa Cum Laude, Top 1/106)

Mar. 2018 – Feb. 2024

## Publication (◦ Subject / ◦ Contribution)

🎓 [Google Scholar](#)

### [1] A11yn: Aligning LLMs for Accessible Web UI Code Generation (COLM 2026) [pdf](#) [🔗](#)

Janghan Yoon, Jaegwan Cho, Junhyeok Kim, Jiwan Chung, **Jaehyun Jeon**, Seungwon Lim, Youngjae Yu

- Developed a GRPO-based framework to align code-generating LLMs with web UI accessibility standards
- Built multi-GPU training pipeline with axe-core-based accessibility compliance reward function design and synthetic datasets

### [2] Do MLLMs Capture How Interfaces Guide User Behavior? A Benchmark for Multimodal UI/UX Design Understanding (ACL 2026 Main) [pdf](#) [🔗](#) [code](#) [🔗](#)

**Jaehyun Jeon**, Min Soo Kim, Jang Han Yoon, Sumin Shim, Yejin Choi, Hanbin Kim, Dae Hyun Kim<sup>†</sup>, Youngjae Yu<sup>†</sup>

- Proposed a multimodal LLM evaluation task for understanding and reasoning about human behavior in UI/UX design
- Built a benchmark using real-world A/B-tested UI pairs with rationales, revealing significant MLLM limitations including near-random selection performance, and outlined future directions for behavior-aware UI generation

### [3] Zero-shot Multimodal Document Retrieval via Cross-modal Question Generation (EMNLP 2025 Main) [pdf](#) [🔗](#)

Yejin Choi, Jaewoo Park, Janghan Yoon, Saejin Kim, **Jaehyun Jeon**, Youngjae Yu

- Developed a multimodal retrieval framework achieving SOTA performance on out-of-distribution benchmarks

- Proposed preQ generation method and demonstrated improved recall on multilingual Chinese and Korean RAG benchmarks
- [4] **Can Visual Language Models Resolve Textual Ambiguity with Visual Cues? Let Visual Puns Tell You!** (*EMNLP 2024 Main*) [pdf](#) [🔗](#)
- Jiwan Chung, Seungwon Lim, **Jaehyun Jeon**, Seungbeen Lee, Youngjae Yu
- Introduced a benchmark evaluating multimodal literacy in models using images for pun understanding and word disambiguation
  - Developed multimodal pun benchmark through image generation/retrieval and conducted model performance evaluations

**Project** (◦ Subject / ◦ Contribution)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>APEX: AI-Powered Patent Specification Drafting Platform via Multi-Agent Pipeline</b>                                                                                                                                                                                                                                                                                                                                                                                                 | Apr. 2026 - Sep. 2026 |
| <ul style="list-style-type: none"> <li>◦ Built a platform for automated patent specification drafting using a multi-agent pipeline with parallel subagents across claim drafting, drawing labeling, and iterative spec generation with escalation logic</li> <li>◦ Led full-stack and model development, and collaborated iteratively with patent attorneys to define domain-specific rubrics as shared evaluation criteria for human review and model self-assessment loops</li> </ul> |                       |
| <b>[Yanolja] Automated Accommodation Image Assessment Based on User Preferences</b> ( <i>Lead</i> )                                                                                                                                                                                                                                                                                                                                                                                     | Jun. 2025 - Aug. 2025 |
| <ul style="list-style-type: none"> <li>◦ Developed an end-to-end framework for assessing accommodation images based on human-perceived attractiveness</li> <li>◦ Built assessment framework identifying informativeness and aesthetics as crucial aspects, created manually labeled datasets, trained and evaluated VLMs, and proposed image sorting strategies to enhance user experience</li> </ul>                                                                                   |                       |
| <b>[LG Electronics] webOS Platform Development Support Multimodal Chatbot</b>                                                                                                                                                                                                                                                                                                                                                                                                           | May. 2024 - Dec. 2024 |
| <ul style="list-style-type: none"> <li>◦ Built RAG-based multimodal chatbot with webOS knowledge base, automating data preprocessing to answering pipeline</li> <li>◦ Designed document preprocessing pipeline with markdown conversion, built curated benchmark for performance analysis, implemented reranking to improve in answer quality, and delivered API for usability</li> </ul>                                                                                               |                       |
| <b>AkaLlama: Korean Specialized 70B LLM Construction</b> <a href="#">link</a> <a href="#">🔗</a>                                                                                                                                                                                                                                                                                                                                                                                         | May. 2024 - Dec. 2024 |
| <ul style="list-style-type: none"> <li>◦ Released a 70B LLaMA-based LLM full fine-tuned with curated data to be specialized in Korean</li> <li>◦ Curated Korean training dataset gathering open-source instruction-following samples through quality analysis and deduplication, and evaluated performance on major Korean benchmarks showing results comparable to leading models</li> </ul>                                                                                           |                       |
| <b>Visual Distractor: Vision-Language Models Fooled by Images</b> ( <i>Lead</i> )                                                                                                                                                                                                                                                                                                                                                                                                       | Sep. 2023 - Dec. 2023 |
| <ul style="list-style-type: none"> <li>◦ Proposed a benchmark measuring VLM hallucination vulnerabilities from visual interference in text question answering</li> <li>◦ Built datasets by generating diverse questions on anomalous synthetic images and evaluated VLMs for incorrect responses</li> </ul>                                                                                                                                                                             |                       |
| <b>User-interactive Image Captioning with Constrained Decoding</b> ( <i>Lead</i> ) <a href="#">code</a> <a href="#">🔗</a>                                                                                                                                                                                                                                                                                                                                                               | Mar. 2023 - Jun. 2023 |
| <ul style="list-style-type: none"> <li>◦ Developed an interactive image captioning system enabling users to fix specific words in the generated captions</li> <li>◦ Implemented constrained decoding for VLMs by modifying Transformers code to enforce fixed words during generation</li> </ul>                                                                                                                                                                                        |                       |
| <b>[Freewheelin] Establishment of Student Performance Indicators in Service</b> ( <i>Lead</i> )                                                                                                                                                                                                                                                                                                                                                                                         | Oct. 2022 - Dec. 2022 |
| <ul style="list-style-type: none"> <li>◦ Identified students' actual math skill levels to highlight the value of the company's service</li> <li>◦ Developed an embedding-based model predicting student math performance, enabling personalized study recommendations</li> </ul>                                                                                                                                                                                                        |                       |

**Awards and Honors**

|                                                                                               |                       |
|-----------------------------------------------------------------------------------------------|-----------------------|
| <b>Scholarship</b>                                                                            |                       |
| Yonsei-to-Yonsei (Y2Y) Scholarship (Full, merit-based)                                        | Mar. 2024 – Feb. 2026 |
| National Science & Technology Scholarship (Full, merit-based)                                 | Feb. 2022 - Feb. 2024 |
| Academic Excellence Scholarship (Full, merit-based)                                           | Sep. 2018 - Feb. 2022 |
| <b>Awards</b>                                                                                 |                       |
| Software Capstone Design Project, Excellence Award (Fall Semester)                            | Dec. 2023             |
| Software Capstone Design Project, Excellence Award (Spring Semester)                          | Jun. 2023             |
| KPMG Ideathon Challenge, 3rd Place (Subj: 'Shopping Service with Carbon Emission Prediction') | Feb. 2022             |
| ICPC Sinchon Summer Algorithm Camp Contest, 2nd Place                                         | Aug. 2021             |

**Teaching**

|                                                           |                       |
|-----------------------------------------------------------|-----------------------|
| <b>Teaching Assistant</b>                                 |                       |
| CSI4101, 4102: Software Capstone Design                   | Sep. 2024 – Dec. 2024 |
| <b>Mentoring</b>                                          |                       |
| YDMC (Guiding high school students in career exploration) | Sep. 2023 – Jun. 2024 |

## Skills

---

### Programming

[Language] Python, SQL, C++, JavaScript

[ML/DL & AI] PyTorch, Transformers (Huggingface), vLLM, LangChain

[Tools & Platforms] Linux, AWS, Docker, Git, WandB, MLflow, Figma

### Language

Korean (Native), English (Fluent, 1.5 years working with US Army, TOEIC 950, OPIc IH)